

Technology Stack

Date	15 July 2026
Team ID	xxxxxx
Project Name	StudyAI – Smart AI educate Companion & Quiz Generator
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>React.js, TypeScript, HTML5, CSS3, Tailwind CSS, Vite</i>	<i>Chosen for building a responsive and interactive user interface with reusable components, fast development, and better performance.</i>

2	Backend / Server-Side	<i>Python Flask</i>	<i>Provides lightweight and fast API development, easy integration with AI services, and efficient handling of file processing and content generation.</i>
3	Database / Data Storage	<i>Supabase (PostgreSQL)</i>	<i>Offers a secure relational database, authentication services, cloud storage, and real-time data management capabilities.</i>
4	Cloud / Hosting / Deployment	<i>Supabase Cloud and Vercel/Netlify</i>	<i>Provides cost-effective hosting, easy deployment, automatic scaling, and reliable cloud infrastructure.</i>
5	Version Control & CI/CD	<i>Git and GitHub</i>	<i>Enables version control, team collaboration, code backup, issue tracking, and seamless project management.</i>

6	Third-Party APIs / Other Tools	<i>Google Gemini API/Groq API, Supabase Storage, PDF.js, Mammoth</i>	<i>Gemini/Groq APIs provide AI-generated summaries, quizzes, flashcards, and study plans. PDF.js and Mammoth extract text from PDF and DOCX files. Supabase Storage securely stores uploaded documents.</i>